CS2200 — Language, Machines and Computations (theory of computation)

Tier	A
Phase (order)	3
Do this first	CS1200
Estimated effort	12 weeks, ~7 h/week
Unlocks	CS3300 (parsing), CS6850, CS6840

Roadmap

1. DFA/NFA, regular languages, closure, pumping lemma, Myhill-Nerode, minimisation.
2. Regular expressions and the NFA<->RE equivalence.
3. Context-free grammars, PDAs, CNF, CYK, pumping lemma for CFLs.
4. Turing machines, decidability, halting problem, reductions, intro to complexity (P, NP).

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Computer - Theory of Automata, Formal Languages and Computation (42 videos). NPTEL *Theory of Automata, Formal Languages and Computation* (Kamala Krithivasan, IIT Madras).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Theory of Computation_Prof. Subrahmanyam Kalyanasundaram (70 videos). NPTEL-NOC IITM *Theory of Computation* (Subrahmanyam Kalyanasundaram, IIT Hyderabad): newer, cleaner.
- MIT 18.404J Theory of Computation, Fall 2020 (25 videos). MIT 18.404J (Fall 2020, Michael Sipser): taught by the author of the standard textbook.
- Theory of Computation (Raghunath Tewari, IIT Kanpur, 2016) (42 videos). Raghunath Tewari (IIT Kanpur, 2016) — the NPTEL ToC course, uploaded by a TCS channel.

⚡ Short videos for intuition and revision

- Theory of Computation & Automata Theory (100+ videos). Neso Academy *Theory of Computation*: one short video per construction (DFA/NFA/PDA/TM).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	8–11	DFA, NFA, epsilon-NFA, subset construction. Convert 5 NFAs to DFAs by hand; simulate a DFA in Python.
2	12–15	Regular expressions <-> automata, regular grammars. Write a regex-to-NFA (Thompson) converter.
3	16–19	Pumping lemma, Myhill-Nerode, DFA minimisation, Moore/Mealy machines. Prove 5 languages non-regular.
4	1–7	Grammars, CFGs, ambiguity, simplification, Chomsky and Greibach normal forms. Convert 3 grammars to CNF.
5	20–25	Pushdown automata, CFG <-> PDA, pumping lemma for CFLs (16). Implement CYK parsing in Python.
6-7	26–30	Turing machines and their variants. Write TMs in a simulator (turingmachine.io) for a^n b^n c^n and binary increment.
8	31–35	Decidability, universal TM, Rice's theorem, Post's correspondence problem. Then Sipser ch. 4-5 + MIT 18.404 (alt) lectures on reductions.
9	36–37	NP-completeness and Cook's theorem. 38-42 (L-systems, DNA computing) are optional.

▼ **All lectures in *Computer - Theory of Automata, Formal Languages and Computation* (42 videos)**

1. Mod-01 Lec-01 GRAMMARS AND NATURAL LANGUAGE PROCESSING
2. Mod-01 Lec-02 GRAMMARS AND LANGUAGES GENERATED
3. Mod-01 Lec-03 GRAMMARS AND LANGUAGES GENERATED (Contd)
4. Mod-01 Lec-04 AMBIGUITY IN CFG
5. Mod-01 Lec-05 SIMPLICATION OF CFG
6. Mod-01 Lec-06 REMOVAL OF UNIT PRODUCTIONS , CHOMSKY NORMAL FORM FOR CFG
7. Mod-01 Lec-07 GREIBACH NORMAL FORM FOR CFG
8. Mod-02 Lec-08 FINAL STATE AUTOMATA
9. Mod-02 Lec-09 NON-DETERMINISTIC FSA
10. Mod-02 Lec-10 NON DETERMINISTIC FSA (Contd)
11. Mod-02 Lec-11 NON DETERMINISTIC FSA WITH E(Epsilon)- MOVES
12. Mod-02 Lec-12 EQUIVALENCE BETWEEN FSA AND TYPE 3 GRAMMARS
13. Mod-02 Lec-13 REGULAR EXPRESSIONS , REGULAR EXPRESSIONS TO NFSA
14. Mod-02 Lec-14 DFSA TO REGULAR EXPRESSIONS
15. Mod-02 Lec-15 PROBLEMS AND SOLUTIONS
16. Mod-02 Lec-16 PUMPING LEMMAS FOR REGULAR SETS AND CFL
17. Mod-02 Lec-17 MYHILL-NERODE THEOREM
18. Mod-02 Lec-18 MINIMIZATION OF DFSA
19. Mod-02 Lec-19 FSA WITH OUTPUT MOORE AND MEALY MACHINES
20. Mod-03 Lec-20 PUSHDOWN AUTOMATA

21. Mod-03 Lec-21 PUSHDOWN AUTOMATA,EQUIVALENCE BETWEEN ACCEPTANCE BY EMPTY STORE
22. Mod-03 Lec-22 PUSHDOWN AUTOMATA CFG TO PDA
23. Mod-04 Lec-23 PUSHDOWN AUTOMATA PDA TO CFG
24. Mod-04 Lec-24 PROBLEMS AND SOLUTIONS-I
25. Mod-04 Lec-25 PROBLEMS AND SOLUTIONS - III
26. Mod-05 Lec-26 TURING MACHINES
27. Mod-05 Lec-27 TURING MACHINES (Contd)
28. Mod-05 Lec-28 TURING MACHINE AS ACCEPTOR , TECHNIQUES FOR TM CONSTRUCTION
29. Mod-05 Lec-29 GENERALIZED VERSIONS OF TURING MACHINES
30. Mod-05 Lec-30 TURING MACHINE AS A GENERATING DEVICE
31. Mod-06 Lec-31 RECURSIVE SETS , RECURSIVELY INNUMERABLE SETS , ENCODING OF TM , HALTING PROBLEM
32. Mod-06 Lec-32 PROBLEMS AND INSTANCES , UNIVERSAL TM , DECIDABILITY
33. Mod-06 Lec-33 RICE'S THEOREM,LINEAR BOUNDED AUTOMATA,PROPERTIES OF TM
34. Mod-06 Lec-34 POST'S CORRESPONDENCE PROBLEMS
35. Mod-07 Lec-35 POST'S CORRESPONDENCE PROBLEMS (Contd) TIME AND TAPE COMPLEXITY OT TM
36. Mod-07 Lec-36 NP - COMPLETE PROBLEMS , COOK'S THEOREM
37. Mod-07 Lec-37 NP - COMPLETE PROBLEMS (Contd)
38. Mod-08 Lec-38 REGULATED REWRITING
39. Mod-08 Lec-39 L - SYSTEMS
40. Mod-08 Lec-40 GRAMMAR SYSTEMS
41. Mod-09 Lec-41 DNA COMPUTING
42. Mod-09 Lec-42 MEMBRANE COMPUTING

Books, notes and websites

- ✅ NPTEL *Theory of Computation* (Raghunath Tewari, IIT Kanpur) — see classcentral.com/mooc/6701/nptel-theory-of-computation
- Book: Michael Sipser, *Introduction to the Theory of Computation*
- Jeff Erickson, *Models of Computation* notes (free)
- JFLAP to simulate automata

Practice / proof of understanding

Build a DFA minimiser and a CYK parser in code; prove 5 languages non-regular with the pumping lemma; reduce 3 problems to Halting.

What your IITM course sheet says

From the PDF: CS2200

Official description: To introduce students to the basic concepts in theoretical computer science, and the formal relationships among machines, languages, grammars, and computational problems. After completing the course, the student will be able to: • Model, compare and analyse different computational models using combinatorial methods. • Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata. • Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation. • Identify limitations of some computational models and possible methods of proving them. • Have an overview of how the theoretical study in this course is applicable to and engineering application like designing the compilers.

Official syllabus (trimmed):

> Four basic themes (but related in a flow) : 44L + 12T Finite Automata & Regular Languages - 13L + 3T Languages vs Problems. Finite State Automata, Regular Languages. Closure properties, Limitations, Pumping Lemma, Myhill-Nerode relations, Quotient Construction. Minimization Algorithm. Non-determinism & Regular Expressions - 8L + 2T Notion of non-determinism. Acceptance condition. Subset construction. Pattern matching and regular expressions. Regular Expressions and Regular languages. More closure properties of regular languages. Grammars & Context-free Languages(CFLs) - 15L + 5T Grammars and Chomsky Hierarchy, CFLs, Regular Grammars, Chomsky Normal Form, Pumping Lemma for CFLs, Inherent Ambiguity of Context-Free Languages, Cock-Younger-Kasami Algorithm, Applications to Parsing. Pushdown Automata(PDA), PDA vs CFLs. Deterministic CFLs. Turing Machines & Computability - 9L + 2T Introduction to Turing Machines, Configurations, Halting vs Looping. Multi-tape Turing machines. Recursive and Recursively enumerable languages. Undecidability of Halting Problem. Reductions. Introduction to Theory of NP-completeness.

Official textbooks: 1. Michael Sipser “Introduction to the Theory of Computation”, 3rd Edition, CENGAGE Learning, 2012. 2. Dexter Kozen “Automata and Computability” Springer-Verlag, 1997.

Official references: 1. J. E. Hopcroft, R. Motwani and J. D. Ullman, "Introduction to automata theory, languages and computation", Pearson Education Asia, 2001 2. H. R. Lewis and C.H. papadinitriou, "Elements of the theory of computation", Prentice Hall International Editions, 1981.